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MS4092 COMPREHENSIVE EXAMINATION

Manufacturing Processes

Semester II 2021/2022

Saturday, 4th March 2023, (27 minutes)

Closed Book and Closed Note

Answer all the questions on this problem sheet. State your answer by circling the letter in front of the most correct/accurate answer. Correct answers get additional scores, while wrong answers or questions that are not answered get zero score. If necessary, you may use the back page of the question sheet as 'scratch paper'. Don't forget to fill in your name, student identity number, and/or your signature at the top of EVERY page (in the space provided).

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1. Fused Deposition Modelling is one type of additive manufacturing process. It heats the filament and pushes it through a nozzle without melting it. To perform this process, the filament should be a ... material
   a. Thermoset
   b. Thermoplastic
   c. Plasticizer
   d. Colorant

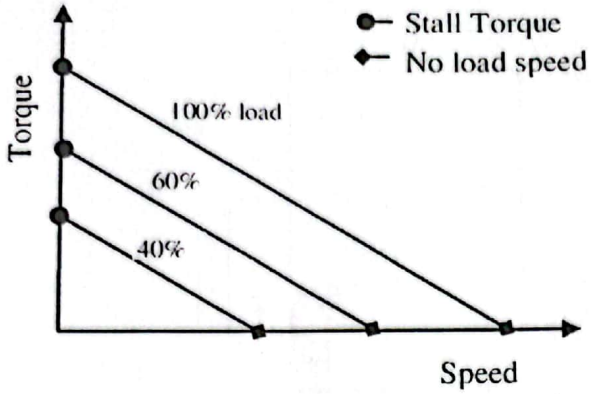
2. A value of how close a measurement is to the true or accepted value is the definition of
   a. Resolution
   b. Accuracy
   c. Precision → kebenaran
   d. Sensitivity

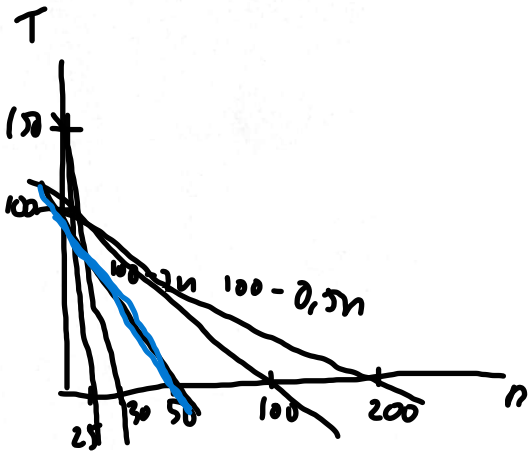
3. Which of the following statements is true about tolerance?
   a. The tolerances of a component are calculated based on the availability of the machine tools in the workshop
   b. The technician in the manufacturing workshop determines the tolerances of a component
   c. The designer determines the tolerances of a component depending on the functions of the parts
   d. The tolerances of the components are determined according to the agreement between the technicians and the designers.

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4.





The picture above shows a DC motor's typical torque–speed characteristics. As a designer, you have a task to choose the most suitable permanent magnet dc motor for your application. Your application requires a torque of 100 N.mm with an angular velocity of 50 rad/s. The relationship between the torque and speed of your motor that you should choose is (n in rad/s)

power design $100n + 50T = 5000$

$T = 100 - 2n$

a. T = 100 <N.mm> - 0.5n <rad/s>
b. T = 100 <N.mm> - n <rad/s>
c. T = 150 <N.mm> - 5n <rad/s>
d. T = 150 <N.mm> - 4n <rad/s>

5. The following statements are the advantages of casting processes, **except**
   a. Produce a part with large dimensions
   b. Casting processes are economics
   c. Produce a part with high accuracy
   d. Produce a part with complex geometries

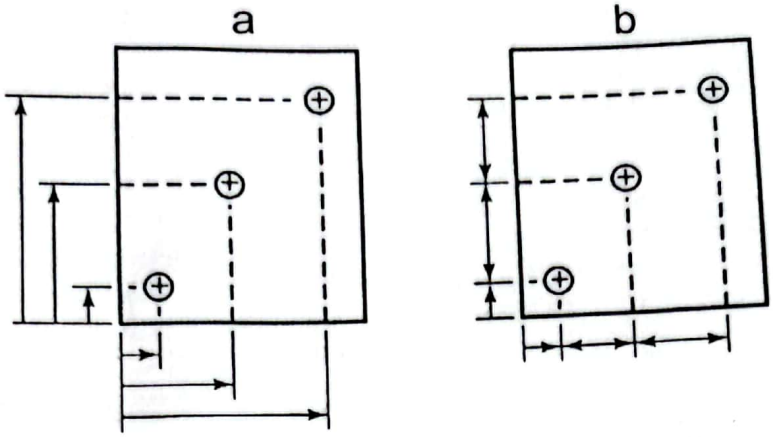
6. A bolt manufacturer produces hexagonal bolts with a diameter of 5 mm with a production rate of 10,000 bolts per day. The suitable processes to achieve this production rate is
   a. The manufacturer should use a lathe machine to manufacture the thread
   b. The manufacturer should use an extrusion process to produce the thread and a casting process to produce the hexagonal head
   c. The manufacturer should use the rolling process to produce the thread and a forging process to produce the hexagonal head
   d. The manufacturing should use the casting process

7. A dial indicator is classified as
   a. A linear measurement instrument with direct reading
   b. A linear measurement instrument with indirect reading
   c. A gage
   d. A Comparative length instrument

8. Which of the following statements about these two figures is true

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absolute incremental

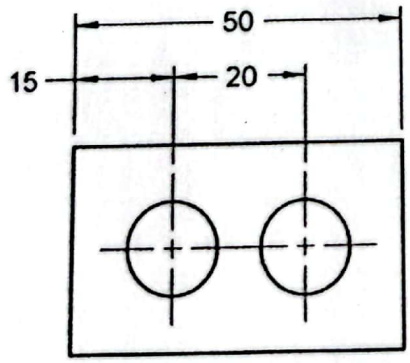


a. Figure A and B show two different referencing methods during the manufacturing process
b. There is no difference between these two figures
c. Dimensioning of Figure B is wrong since the dimension should start on the edge
d. Figure A is called incremental dimensioning, and Figure B is called absolute dimensioning

9. While performing a measurement, we need to avoid parallax errors.
   a. Parallax error happens if the line of sight is perpendicular to the measurement arrow and the marking
   b. Parallax error can be avoided if the line of sight is perpendicular to the measurement arrow and the marking
   c. Parallax can be avoided if the measurement arrow is made from a very thin material
   d. Parallax error can be avoided by performing the measurements in a room with enough light intensity

10. Which of the following processes could produce the narrowest tolerance range?
   a. Hot extrusion
   b. Investment casting
   c. Turning
   d. Polishing

11.



What can you conclude from the above drawing?
   a. The distance between the two holes is more important than the distance between the right hole to the left side of the part

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b. The distance between the right hole to the left side of the part is more important than the distance between the two holes
c. There is no underlying meaning in the drawing. The distance from the right hole to the left side of the part is 35 mm
d. There is a mistake in the drawing. The distance between the right hole and the part's right side should also be stated.

12. Based on your experience as a mechanical engineering student, which of the following statement is true while designing a product so that it will be easier to assembly
a. During the assembly, parts can be inserted from any directions
b. Increase the number and variety of parts in a product
c. A high degree of asymmetry in the parts
d. Physical obstructions to the free movements of the parts during assembly are not a concern

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